Kiron Sukesan

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Summary

Research assistant in Wei Lu's group at the University of Michigan (EECS), testing a state-space model of hippocampal time cells against neural recordings. Previously nine years in industry: production ML and computer vision at Compass, then distributed storage at AWS, including two years managing the DynamoDB Storage Node team. Research background in Bayesian active learning and closed-loop BCIs, with three peer-reviewed publications. Applying to PhD programs for Fall 2027 to work on understanding what large models represent internally and how multiple AI systems behave together.

Skills

Languages	Python, Java, C++, Rust, JavaScript, MATLAB, Bash
ML / AI	PyTorch, Deep Learning, Reinforcement Learning, Computer Vision, Gaussian Processes, Bayesian Inference, Semi-Supervised Learning, ANN/HNSW
Storage Systems	Distributed Database Internals, Consensus Protocols (Paxos, Raft), Database Engine Implementation (LSM tree and B-tree)

Experience

Research Assistant *at* **Wei Lu Group, EECS (ECE), University of Michigan** Oct 2026 – Present
Ann Arbor, MI

- Testing a state-space model of hippocampal time cells (Lu et al., arXiv 2507.13638) against calcium-imaging recordings of mouse CA1, asking whether the model's time-cell-like units match recorded neurons
- Building the model-to-data comparison pipeline in Python: matched task environment, the same time-cell classifier applied to model units and recorded neurons, and a similarity metric compared across initializations (HiPPO, random, real-diagonal)

Lead Engineer *at* **Freelance** Apr 2026 – Present
Ann Arbor, MI

- Architected and developed AppMan, a full-stack apartment management platform using Angular, PostgreSQL, and Docker, modernizing legacy Oracle APEX workflows for property operations, payments, leasing, taxes, and task management. In production at a 1,000+-unit property management company.
- Designed and built a test-driven engineering infrastructure with unit, integration, adversarial, migration, authorization, and smoke tests.

Software Development Manager *at* **Amazon Web Services (AWS)** Aug 2024 – Apr 2026
DynamoDB Storage Node, Seattle, WA

- One of the engineering leaders for DynamoDB's Storage Node, among AWS's highest-scale distributed storage systems, overseeing architecture, delivery, and team health. Specifically owned the TTL system, replication protocol, general correctness of DynamoDB Storage, and feature expansion efforts within storage.
- Drove hiring, mentoring, and performance management for a team of 10+ junior, mid-level and senior software engineers; established engineering standards and oncall culture.

Senior Software Engineer *at* **Amazon Web Services (AWS)** Jan 2023 – Aug 2024
DynamoDB Storage Node API Team, Seattle, WA

- Designed and implemented features for the DynamoDB Storage Node API layer, contributing to a system serving millions of customers with stringent latency, availability, and durability requirements.
- Led oncall and operational excellence initiatives, driving root cause analyses and systemic improvements to reduce customer-impacting events across the storage tier.

Machine Learning Engineer II / Senior Machine Learning Engineer *at* **Compass** Dec 2019 – Nov 2022
Seattle, WA

- Built and owned computer-vision-based ML models used by real estate agents across the platform, directly influencing transaction outcomes.
- Architected end-to-end ML pipelines spanning data ingestion, feature engineering, model training, evaluation, and production deployment on cloud infrastructure.

Software Development Engineer I/II *at* **Amazon Kindle** Oct 2017 – Nov 2019
Greater Seattle Area

- Designed and shipped features for Amazon Kindle's internal data processing pipeline, collaborating with senior engineers on distributed systems design, API contracts, and service reliability.
- Promoted to SDE II after demonstrating technical ability with an end-to-end solution for Kindle compliance with GDPR's requirement to return data to customers on request.

Data Scientist / Research Engineer *at* **Washington University in St. Louis** May 2016 – Aug 2017
Laboratory of Sensory Neuroscience and Neuroengineering, St. Louis, MO

- Developed a Bayesian active learning framework using Gaussian Process Classification to efficiently estimate audiometric threshold surfaces, reducing clinical measurement burden while preserving diagnostic accuracy.
- Designed a novel ML-based hardware calibration algorithm for audiometric equipment, improving signal fidelity and measurement consistency across experimental sessions.
- Co-authored three peer-reviewed publications on Bayesian active learning, psychometric field estimation, and online machine-learning audiometry.

Software Developer *at* **Purnatva Solutions Pvt. Ltd.** Sep 2014 – Aug 2015
Bengaluru, India

- Implemented real-time GPS bus tracking module for the Falcon Mobile App, supporting large-scale deployment with 10,000+ active users across the platform.

Biomedical Engineer, Undergraduate Research *at* **Wash. U. in St. Louis** Aug 2013 – May 2014
Department of Biomedical Engineering

- Designed and built a wearable EEG recording device for overnight clinical use, using custom analog circuitry, embedded software, and mechanical design in close collaboration with two biomedical engineering peers and clinical partners.
- Validated device performance against clinical EEG standards and iterated on hardware and firmware based on clinician feedback.

Neurosurgery Research Intern *at* **University of Washington** Jun 2013 – Aug 2013
Brain-Computer Interface Lab (Dr. Jeff Ojemann), Seattle, WA

- Developed a signal analysis algorithm for electrocorticographic (ECoG) neural signals; diagnosed and resolved electromagnetic noise artifacts that had been corrupting recorded data.
- Constructed a closed-loop BCI device that enabled motor-impaired patient hand movement through real-time classification of theta wave activity from ECoG recordings.

Education

Washington University in St. Louis 2015 – 2016
Master of Science, Computer Science St. Louis, MO

Washington University in St. Louis 2010 – 2014
Bachelor of Science, Biomedical Engineering St. Louis, MO

Publications

- Song, X. D., **Sukesan, K. A.**, & Barbour, D. L. "Bayesian active probabilistic classification for psychometric field estimation." *Attention, Perception, & Psychophysics*, 80 (2018), 798–812. DOI: 10.3758/s13414-017-1460-0.
- Barbour, D. L., DiLorenzo, J. C., **Sukesan, K. A.**, Song, X. D., Chen, J. Y., Degen, E. A., Heisey, K. L., & Garnett, R. "Conjoint psychometric field estimation for bilateral audiometry." *Behavior Research Methods*, 51 (2019), 1271–1285. DOI: 10.3758/s13428-018-1062-3.
- Barbour, D. L., Howard, R. T., Song, X. D., Metzger, N., **Sukesan, K. A.**, et al. "Online machine learning audiometry." *Ear and Hearing*, 40(4) (2019), 918–926. DOI: 10.1097/AUD.0000000000000669.